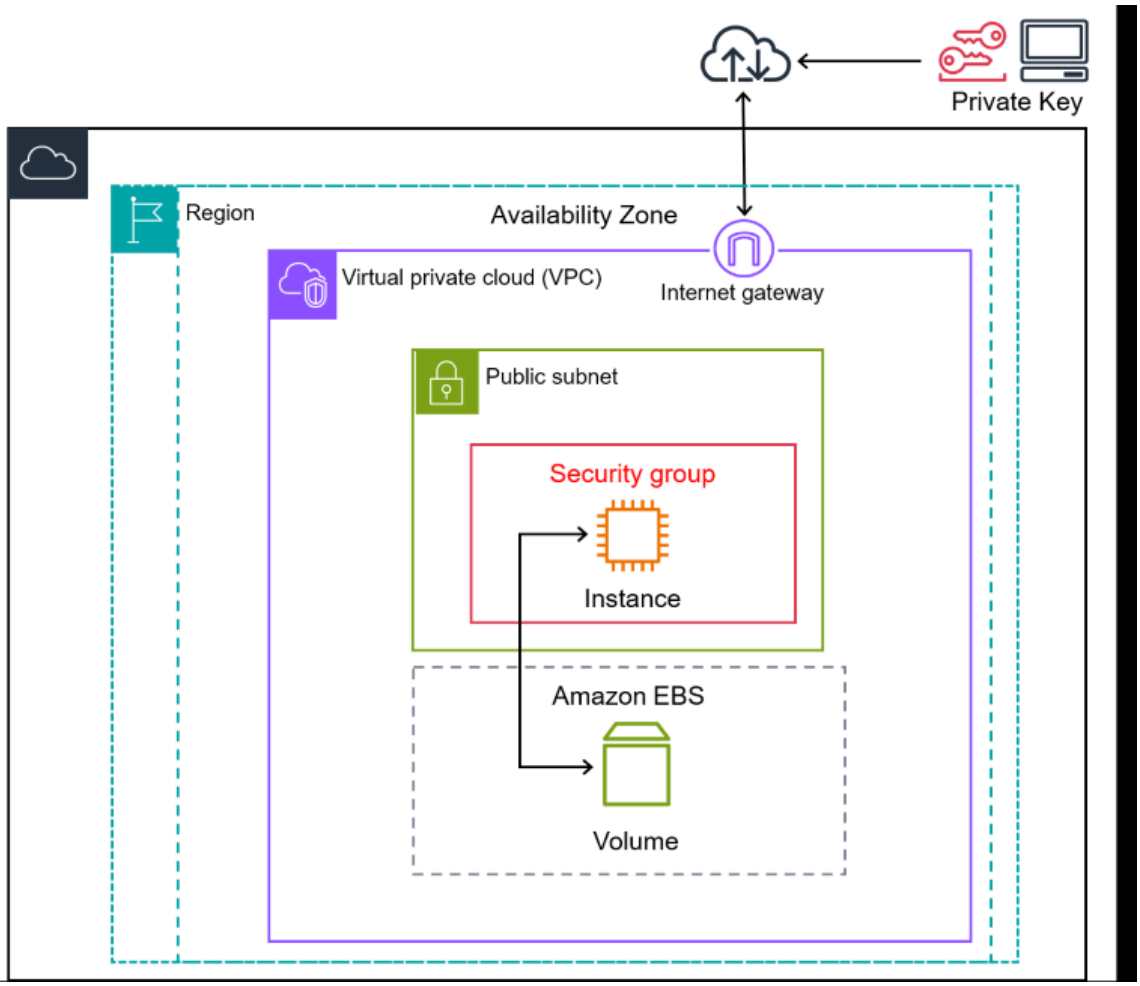
Case study on Amazon EC2 and learn about Amazon EC2 web services

Laboratory Practice-II (Part-2)

- ## What is Amazon EC2?

Amazon Elastic Compute Cloud (Amazon EC2) provides on-demand, scalable computing capacity in the Amazon Web Services (AWS) Cloud. Using Amazon EC2 reduces hardware costs so you can develop and deploy applications faster.



In this example, the EC2 instance is within an Availability Zone in the Region. The EC2 instance is secured with a security group, which is a virtual firewall that controls incoming and outgoing traffic. A private key is stored on the local computer and a

public key is stored on the instance. Both keys are specified as a key pair to prove the identity of the user. In this scenario, the instance is backed by an Amazon EBS volume. The VPC communicates with the internet using an internet gateway.

Features of Amazon EC2

Amazon EC2 provides the following high-level features:

Instances

Virtual servers.

Amazon Machine Images (AMIs)

Preconfigured templates for your instances that package the components you need for your server (including the operating system and additional software).

Instance types

Various configurations of CPU, memory, storage, networking capacity, and graphics hardware for your instances.

Key pairs

Secure login information for your instances. AWS stores the public key and you store the private key in a secure place.

Instance store volumes

Storage volumes for temporary data that is deleted when you stop, hibernate, or terminate your instance.

Amazon EBS volumes

Persistent storage volumes for your data using Amazon Elastic Block Store (Amazon EBS).

Regions, Availability Zones, Local Zones, AWS Outposts, and Wavelength Zones

Multiple physical locations for your resources, such as instances and Amazon EBS volumes.

Security groups

A virtual firewall that allows you to specify the protocols, ports, and source IP ranges that can reach your instances, and the destination IP ranges to which your instances can connect.

Elastic IP addresses

Static IPv4 addresses for dynamic cloud computing.

Tags

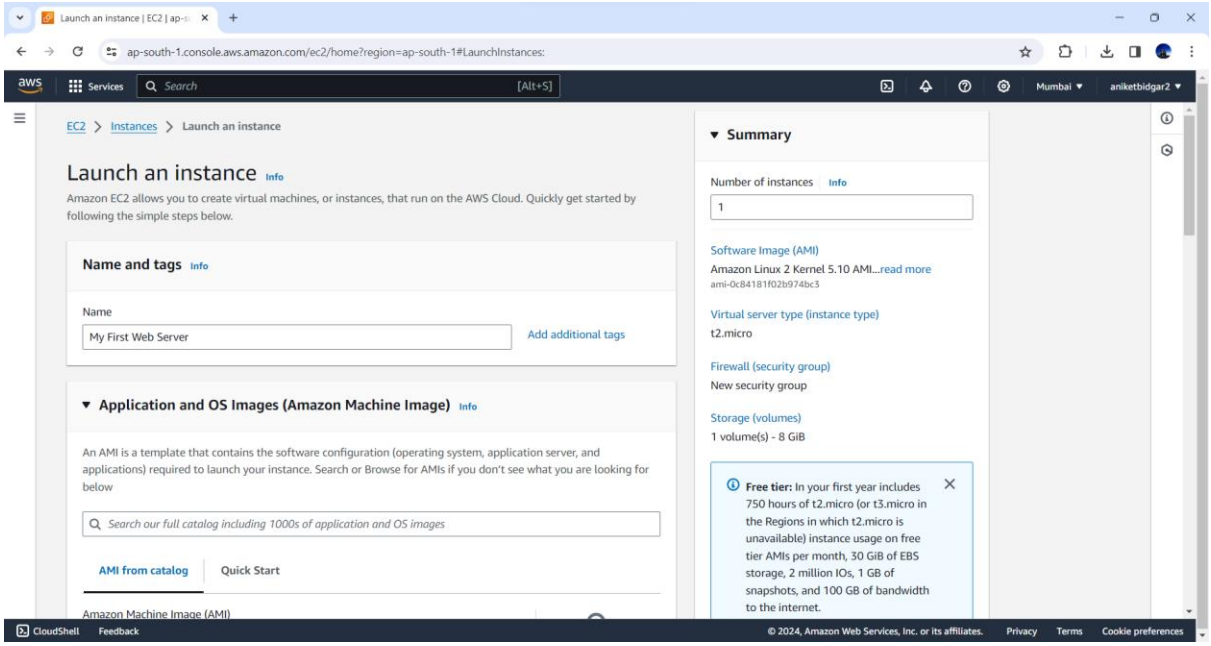
Metadata that you can create and assign to your Amazon EC2 resources.

Virtual private clouds (VPCs)

Virtual networks you can create that are logically isolated from the rest of the AWS Cloud. You can optionally connect these virtual networks to your own network.

To launch an instance

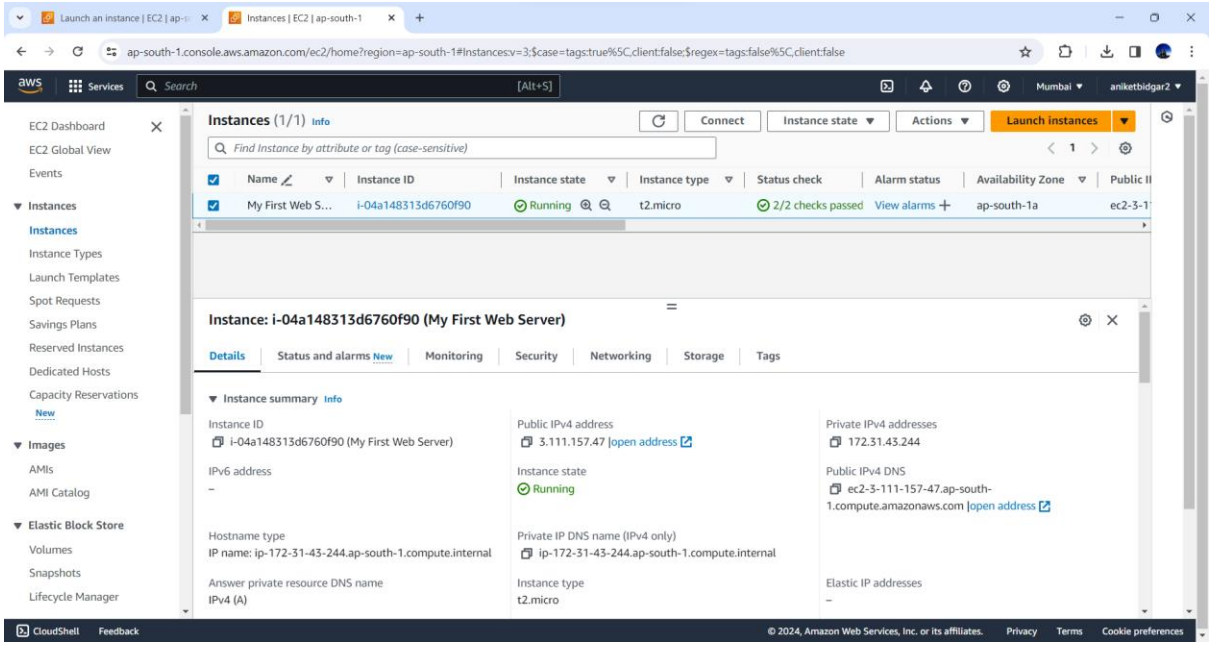
1. Open the Amazon EC2 console at https://console.aws.amazon.com/ec2/.
2. From the EC2 console dashboard, in the Launch instance box, choose Launch instance.
3. Under Name and tags, for Name, enter a descriptive name for your instance.
4. Under Application and OS Images (Amazon Machine Image), do the following:
    a. Choose Quick Start, and then choose Amazon Linux. This is the operating system (OS) for your instance.



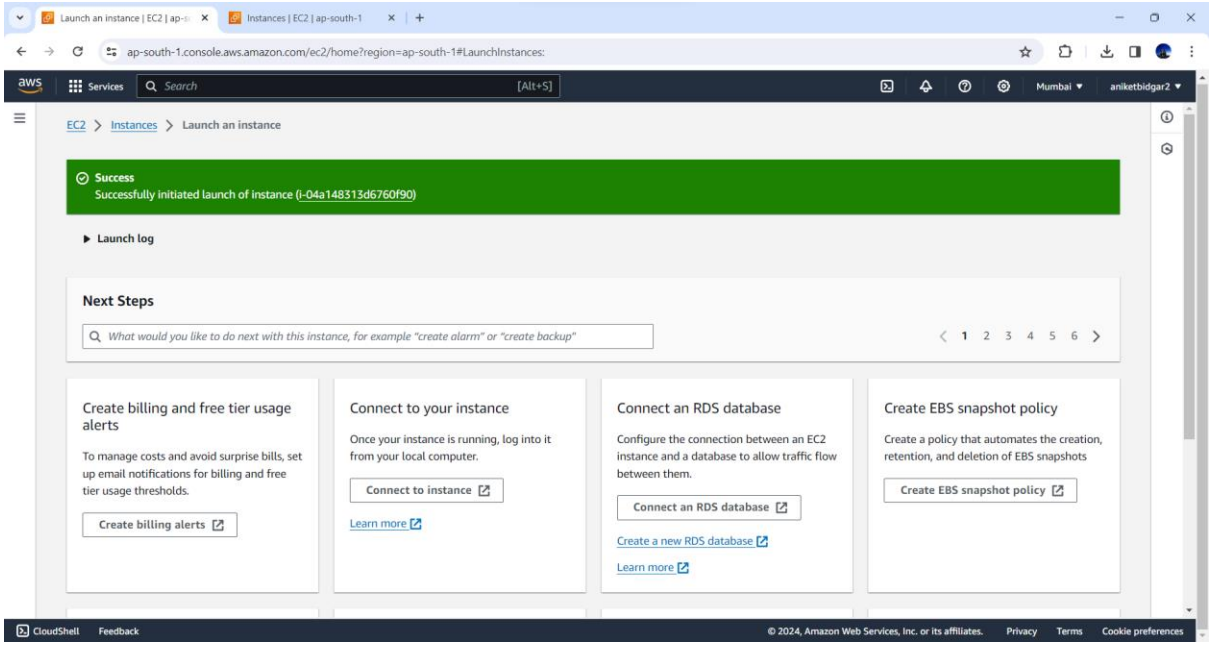
b. From Amazon Machine Image (AMI), select an HVM version of Amazon Linux 2. Notice that these AMIs are marked Free Tier eligible. An *Amazon Machine Image (AMI)* is a basic configuration that serves as a template for your instance.

5. Under Instance type, from the Instance type list, you can select the hardware configuration for your instance. Choose the `t2.micro` instance type, which is selected by default. The `t2.micro` instance type is eligible for the Free Tier. In Regions where `t2.micro` is unavailable, you can use a `t3.micro` instance under the Free Tier.

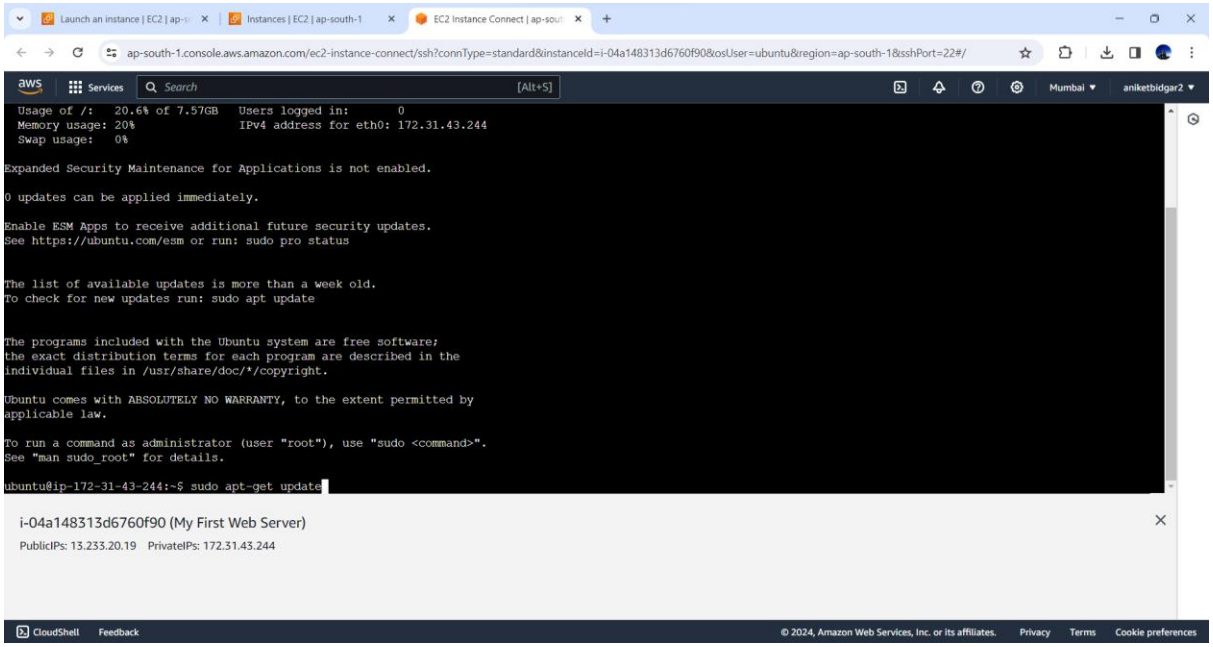
6. Under Key pair (login), for Key pair name, choose the key pair that you created when getting set up.



1. Next to Network settings, choose Edit. For Security group name, you'll see that the wizard created and selected a security group for you. You can use this security group, or alternatively you can select the security group that you created when getting set up using the following steps:
    a. Choose Select existing security group.
    b. From Common security groups, choose your security group from the list of existing security groups.
2. Keep the default selections for the other configuration settings for your instance.
3. Review a summary of your instance configuration in the Summary panel, and when you're ready, choose Launch instance.



1. A confirmation page lets you know that your instance is launching. Choose View all instances to close the confirmation page and return to the console.
2. On the Instances screen, you can view the status of the launch. It takes a short time for an instance to launch. When you launch an instance, its initial state is `pending`. After the instance starts, its state changes to `running` and it receives a public DNS name. If the Public IPv4 DNS column is hidden, choose the settings icon (⚙) in the top-right corner, toggle on Public IPv4 DNS, and choose Confirm.
3. It can take a few minutes for the instance to be ready for you to connect to it. Check that your instance has passed its status checks; you can view this information in the Status check column.


Launch an instance | EC2 | ap-s
Instances | EC2 | ap-south-1
EC2 Instance Connect | ap-south
ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh?connType=standard&instanceId=i-04a148313d6760f90&osUser=ubuntu&region=ap-south-1&sshPort=22#/
aws
Services
Search
[Alt+S]
Mumbai
aniketbidgar2
Usage of /: 20.6% of 7.57GB Users logged in: 0
Memory usage: 20% IPv4 address for eth0: 172.31.43.244
Swap usage: 0%
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
ubuntu@ip-172-31-43-244:~$ sudo apt-get update
i-04a148313d6760f90 (My First Web Server)
PublicIPs: 13.233.20.19 PrivateIPs: 172.31.43.244
CloudShell
Feedback
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